

Ventrilo-wrist: A Wrist Brace Preventing the Onset of Carpal Tunnel Syndrome

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Abstract

Carpal Tunnel Syndrome (CTS) affects around 4.8 million employed adults in the United States, worth a 12-month prevalence of 3.1% for workers aged 18-64. This condition happens when the median nerve is compressed within the carpal tunnel which causes pain, numbness, muscle atrophy, and reduced hand function. Other than the quality-of-life reductions, there are severe economic burdens including an upfront cost of \$85,000 to \$110,000 and a passive \$31,000 dollars every year. Current preventative solutions remain limited, with passive braces offering only symptomatic relief and existing active monitoring systems lacking real-time corrective feedback mechanisms. The pathophysiology of CTS happens based on extreme wrist movements and extreme flexion/extension angles which are able to increase the carpal tunnel pressure from a normal range of 2 to 10 mmHg to over 8x that starts a chain reaction of inflammatory cascades and progressive nerve damage. This project, Ventrilo-Wrist, addresses this need for a low-cost wearable device that continuously monitors wrist posture using Inertial Measurement Units (IMUs). The device was tested across multiple prototyping phases, validating IMU accuracy against goniometer measurements and optimizing the corrective mechanism for user comfort and effectiveness. Results demonstrated successful detection of flexion/extension beyond 15 degrees and radial/ulnar deviation exceeding 10 degrees, with corrective feedback initiated within 200 milliseconds.

Acknowledgements

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Ventrilo-wrist: An Ergonomic Wrist Brace Which Prevents the Onset of Carpal Tunnel Syndrome

Carpal Tunnel Syndrome is a prevalent condition that affects millions in every industry. In fact, 6.7% of employed adults aged 18–64 in the United States reported having carpal tunnel syndrome, with a 12-month prevalence of 3.1%, representing approximately 4.8 million individuals in 2010 (Luckhaupt et al., 2013; Centers for Disease Control and Prevention [CDC], 2010). This number is likely to be higher today due to population growth and modern societies' increasing reliance on repetitive, wrist-intensive tasks. The social and economic impact of CTS is an equally large problem, with employed people who develop serious CTS losing between 85,000 and 110,000 dollars in total, including hospital bills and salary loss. Additionally, workers with CTS miss a median of 138 workdays, nearly three times more than those with fractures. Moreover, they earn around \$31,000 less every decade due to their disability (Ruttenberg, 2019).

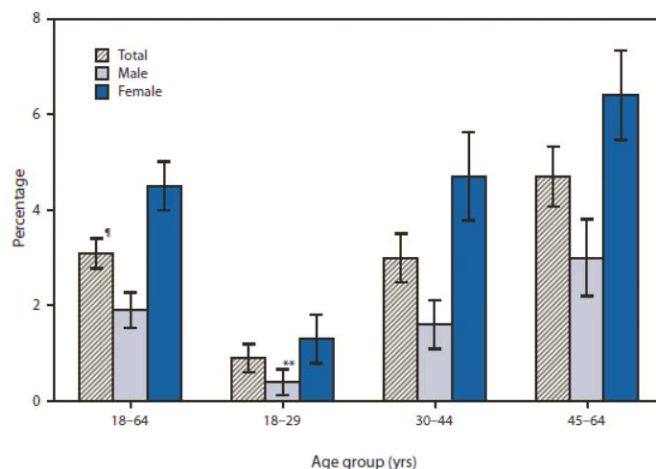


Figure 1

Percentage of employed adults aged 18–64 years who had carpal tunnel syndrome in the past 12 months, by sex and age group — National Health Interview Survey, 2010 (Centers for Disease Control and Prevention, 2010).

There are certain factors that increase an individual's likelihood of developing CTS.

Women experience higher rates of CTS compared to men, at 5.4% and 1.9%, respectively.

Additionally, CTS incidence rises with age, starting at around 0.9% among individuals aged 18–29 and increasing to 4.7% among those aged 45–64. As seen in Figure 1, younger populations

have generally not yet developed CTS, suggesting that preventative measures implemented early may reduce incidence rates over time. Furthermore,

lower education levels correlate with higher CTS risk—those with only a high school diploma show a 4.5% prevalence, nearly double that of individuals with a bachelor's degree (Luckhaupt et al., 2013). The most important factor is the type of occupation, as the nature of one's work and its impact on wrist use plays a decisive role in CTS development, being far more common in manufacturing, office, and administrative roles. In fact, 24% of patients diagnosed with CTS were in the manufacturing industry—a rate 2.53 times higher than the proportion of manufacturing workers in the overall tested population (Luckhaupt et al., 2013). Understanding these demographic

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factors provides context for how internal mechanisms drive the condition, and looking at pathophysiology helps identify the root cause to address.

Pathophysiology

In general, CTS occurs when the median nerve, one of the major nerves of the hand, gets compressed within the carpal tunnel. The carpal tunnel is an osteofibrous conduit formed by the carpal bones and the transverse carpal ligament, which contains nine flexor tendons and the median nerve. This tunnel is at the base of the palm, and as a result, it has a rigid volume, which means that even a small increase in swelling or thickening can raise the pressure significantly. Additionally, there are several important tendons and nerves that pass through the tunnel, and the median nerve provides sensation in the thumb and first two and a half fingers. CTS is frequently associated with fibrous hypertrophy (formation of scar tissue) of the synovial flexor sheath from repeated inflammation or injury. In addition, there are certain risk factors that increase someone's chance of getting CTS: repetitive wrist movements, frequent wrist extension/flexion, constant vibratory exposure, pregnancy, obesity, and hypothyroidism.

Conditions such as diabetes, menopause, and rheumatoid arthritis also contribute to CTS because they can promote inflammation and alter blood flow which increases tunnel pressure. When the median nerve gets compressed, it leads to disturbances in intraneural microcirculation, myelin sheath integrity, axonal tension, and supporting tissues. This process happens very frequently at the proximal edge of the carpal tunnel and the narrowed area near the hook of the hamate, a bone in the hand. Normal pressure is 2-10 mmHg, but when the wrist is in motion, it can rise to 30 mmHg, higher during flexion and extension. This sustained pressure leads to microcirculatory injury—an ischemic vascular process that disrupts the blood-nerve barrier (BNB). Persistent injury and inflammation causes the accumulation of proteins and inflammatory cells which lead to intravascular edema. As the swelling increases, the amount of oxygen that can be delivered to the nerve is reduced, leading to the expression of hypoxia-related molecules that perpetuate inflammation and vascular leakage. Sustained ischemia leads to progressive neurovascular changes such as basement membrane thickening, perineurial fibrosis, demyelination, and nerve fiber degeneration. As neuronal damage becomes chronic, these structural and biochemical changes greatly reduce mobility, general health of the nerve, and wrist movement, and this creates a cycle of chronic pressure and injury. Chronic inflammation around the median nerve leads to reduced blood flow and causes HIF-1 α and VEGF to be regulated. Chronic inflammation around the median nerve can impact the connective tissue, reducing how much the

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nerves can glide and impacting the severity of CTS. Repetitive movements of the wrist can cause stiffness in the synovial tissue and lead to CTS (Aboonq ,2015).

Synovial tissue also plays a major part in how CTS develops over time. Synovial tissue is a key regulator of the biomechanical environment inside the carpal tunnel, as it can lubricate the flexor tendons and help minimize the friction during wrist movements. Glycoproteins and hyaluronic acid are produced by the synovial tissue to help tendons glide smoothly. The key structure for CTS is the subsynovial connective tissue (SSCT) that acts as a buffer, absorbing shear forces and preventing tendon-to-tendon abrasion, thereby absorbing shear force and prevent tendon to tendon abrasion, allowing it to distribute load equally and effectively. As with every part of CTS, chronic wrist overextension, exposure to vibration, and high-force gripping forces disrupt the SSCT structure. These stresses means that it develops microtears that activate fibroblasts that causes the thickening of the SSCT. SSCT thickening reduces the tendon gliding efficiency and leads to increased friction, putting it through this self-perpetuating cycle of friction and inflammation in the tissue. Since the SSCT connects everything together, the inflammation can decrease the space, thereby constricting the space for every other component to fit and thereby constricts the median nerve, leading to CTS. This pressure can increase with wrist deviations, worsening nerve ischemia (Werthel et al., 2014). Because pathological changes are driven by extreme wrist deviations, effective prevention depends on the ability to monitor these movements continuously and actively correct. This type of solution involves using sensors and microcontrollers that can accurately measure wrist position and act accordingly in real time.

Hardware

Sensors

Many competitor solutions incorporate flex sensors, variable resistors that change resistance based on how much they are bent. The bend increases the distance that electrons have to move within the flex sensor which means that there is high resistance. The typical resistance is 25k ohms unbent and only require an ADC (analog to digital converter) input, making them electrically simple to implement. However, the bend angle and resistance do not follow a predictable linear path. They are made thin to allow them to bend, specifically with polymer substrates coated with carbon or conductive ink. A major downside is that they only give one dimension curvature sensing, not allowing for 3D orientations or rotations or twists, a critical part for understanding how a wrist is oriented and moves. These sensors often experience drift and fatigue in a much faster manner, and are structurally fragile. The

use of a flex sensor is best suited for a simplistic prototype, commonly used to test an idea (Agarwal, 2019). As a result, inertial measurement units (IMUs) are better suited for prototypes that require accurate multi-axis motion tracking rather than single-axis curvature sensing. Inertial Measurement Units (IMUs) can be either 6 Degrees of Freedom (DOF) or 9 DOF, having a 3-axis accelerometer and gyroscopes, some of them even having 3-axis magnetometers. They can tell a 3-dimensional kinematic reconstruction of movements and give important benchmarks such as dynamic acceleration, rotational velocity, pitch, roll, and yaw. This can enable flexion, extension, radial, and ulnar deviation alongside pronation and supination. However, utilizing IMU's are much more technically complicated to use and requires several algorithms to be able to take the raw data and turn it into matrices for analysis. Additionally, gyroscope drift does accumulate over time, which can be corrected using alignment with accelerometer gravity alignment. It does require stable mechanical mounting to avoid motion artifacts. There is a tradeoff between ease of use and complexity with faster, more accurate data collection (Shawn, 2019a).

Additionally, EMG sensors can be used as they are able to detect muscle fiber depolarization through voltage changes measured through the skin. It would be able to give more biomechanically relevant data, muscle requirement patterns, fatigue, and activation time, etc. EMG sensors can give a more predictive warning compared to that of angles and they are built in amplification, filtering, and conditioning in modern sensors. It can sense whether there is sustained wrist flexion or extension, but the signal commonly requires preprocessing through rectification, envelope extraction, smoothing, and spectral analysis to reduce noise and improve reliability (Shawn, 2019b; Ayrapetyan et al., 2021). However, EMG sensors can't differentiate between certain muscles or nerves but can only understand aggregate muscle activity. One cannot find the joint angle or orientation, but gives the neuromuscular effort which needs interpretation to be able to analyze it. Finally, it is much more complicated to integrate into systems due to the electrode/pads and if there is any varying texture (Shawn, 2019b). Sensors give the data necessary to identify angles but there is another central piece, the microcontroller. This following section discusses the microcontroller tradeoffs that can support continuous, high-resolution motion tracking.

Micro-Controllers

The Arduino ATmega328P is an 8-bit AVR RISC microcontroller. A single core clocked at 16 MHz means it can only focus on one operation at a time, and it also has significantly limited memory, having a 2 KB SRAM with 32

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KB flash without any external RAM options. It is limited to a 10-bit ADC with limited GPIO, and is used for low-frequency sensors, unlike the requirements for an IMU. Additionally, it does not have an inbuilt external shield but can add modules which expand functionality. The Arduino is designed for suitable, low complexity, and frequency sensing designs, one that would limit the amount of data that is able to be collected (OpenELAB, 2024). The ESP32 Xtensa LX6 SoC has a dual-core 32-bit Xtensa LX6 processor that runs at 240 MHz with 520KB SRAM and external flash. There is integrated Wi-fi and Bluetooth alongside a FPU, which can help support floating-point operations. Due to the dual core, tasks can be split so that the first core can be used to do sensing, and the second core can be used to communicate or actuate (OpenELAB, 2024). The Raspberry Pi Pico is exceptional for precise I/O timing and medium frequency IMU work, but it lacks a hardware floating-point unit (FPU) and robust wireless integration for continuous wearable analytics. For the Raspberry Pi, it is also a dual core, but it is an ARM Cortex-Mo+, which is 133MHz, faster than Arduino but slower than ESP32. It has 264 KB SRAM, and it has a unique PIO, allowing for hardware-level state machines for custom serial protocols and ultra precise timing. It does not include wireless communication unless one specifically acquires the Pico W, something that has weaker wireless capability than ESP32 (OpenELAB, 2024). For the computation throughput, the Arduino is much less than the Pico, which is less than the ESP 32.

Additionally, ESP 32 has a hardware FPU meaning that it does all of its float-pointing calculations much faster than the Pico or Arduino. For the wireless aspect, Arduino relies on external modules; the Pico needs to be a Pico W, while the ESP32 has better Wireless connectivity than both since it has native Wi-Fi and BLE integrated. Arduino being single core means it can only focus on one task at a time, while the ESP32 does have dual core meaning that it can focus on two separate tasks at the same time (OpenELAB, 2024). Since for this project the real time delay will be able to provide the most effective feedback, and it should be able to do computational complicated matrix math in real time alongside wireless communication, ESP32 is the best choice for this project. Using this sensing and hardware framework, it is important to see current progress in the field and understand the strengths and limitations of current designs.

Braces

Passive Braces

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For passive braces, there is Manu® Soft Hand Brace, a soft and compressive brace that wraps around the metacarpals and stabilizes the fingers, changing the carpal tunnel geometry while worn. It is designed to help by reducing the pressure in the median nerve, where 10 participants tested it and reported highly subjective pain reduction while wearing the brace (Manente et al., 2001). Additionally, when the Manu soft brace was compared with a CAMP Tielle® rigid wrist splint, 140 CTS patients were tested using the Boston Carpal Tunnel Questionnaire and Visual Analog Scale and found that both devices provided symptomatic benefit (De Angelis et al., 2009).

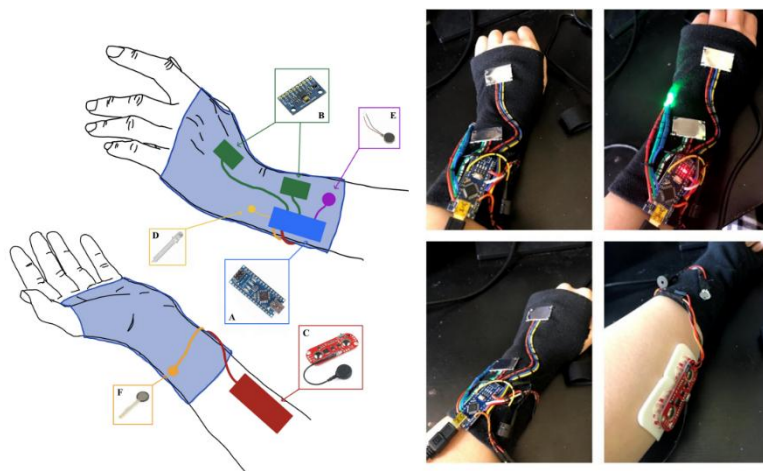
Active Braces

Figure 2 The picture on the left shows the prototype setup with microcontroller (A), two IMUs (B), EMG (C), RGB LED (D), vibration motor (E), and FSR (F). On the right is the physical prototype.

Michael Mack and Cheol-Hong Min developed an active brace that utilized flex sensors calibrated in MATLAB, an Arduino Pro Mini, and a Bluetooth Mate RN-42 to detect hazardous wrist postures through real-time data collection. They suggested improving the angle calibration, adding multi-axis sensing, brace comfort, and real time feedback (Mack & Min, 2019). In

Umar et al. (2019), they utilize an IMU (verified using goniometer measurements with 32 people) to give real-time, consistent measurements as a tool for ergonomic measurement and not for patient treatment. Future work would center around more extensive field testing, having sensor fusion algorithms, and integrated automatic ergonomic scoring (Umar et al., 2019). In In Ayrapetyan et al. (2021)., they use two IMUs (in the wrist and forearm) and an EMG to calculate an overuse score which triggers real-time alerts, as shown in figure 2. In figure 2, the author shows the location of all the sensors and the wiring, helping illustrate how they use IMUs and EMGs physical locations to detect overuse. For the material, they used conductive fabric both for user comfort and simplicity, focusing on preventing CTS symptoms rather than curing the syndrome. Improvements include improving the EMG signal stability, adding more user tailored thresholds, and developing it to become more durable in the long term

(Ayrapetyan et al., 2021). Piper (2017) used a multisegmented wrist brace designed using additive manufacturing with rotary encoders and Force Sensing Resistors. Additionally, they tested it by using VICON to differentiate between certain tasks based on angle and velocity, helping them between different ergonomic assessment frameworks. The researcher believed that they could have expanded to larger subject populations, adding force sensors, improve comfort and weight distribution, and integrate wireless communication (Piper, 2017).

Problem Statement(s)

Problem 1: CTS develops gradually because of prolonged exposure to harmful wrist positions, yet people often don't know when their wrist positions increase the pressure in the Carpal Tunnel which causes CTS. Although there is a clear biomechanical relationship between the wrist deviations and median nerve compression/damage, there is no widely available commercial device that can provide continuous real time awareness of wrist posture.

Problem 2: Research about preventative solutions for CTS that provide real time feedback or promote behavioral change are only able to detect the posture without being able to correct the user in real time actively. As a result, there is a gap in wearable technology which combines accurate multi-axis wrist posture monitoring with active correction to guide users away from hazardous positions.

Objective

Obj. 1a: Design and implement a wearable system that can continuously monitor multi-axis wrist posture in real time using inertial measurement units.

Obj. 1b: Confirm the precision and dependability of the measures for wrist posture by comparing angles measured through sensors with quantitative statistical analysis against reference measures

Obj. 2a: Design an active corrective mechanism that responds to hazardous wrist angles by guiding the wrist toward a neutral position.

Obj. 2b: Integrating sensing, correction, and processing components into an ergonomic, wearable wrist brace suitable for continuous use and preventative intervention.

Section II: Methodology

1. Opening Paragraph (Role of Student vs. Mentor)

All engineering design, prototyping, code development, experiment configuration, data acquisition, statistical analysis, and graph development were conducted within a span of five months. Though conceptual advice was available from mentors, all code, computational, and analytical work was executed independently.

2. Materials and Equipment

ESP32-S3 Microcontroller (refer to Appendix B2 for justification), Two MPU6050 6-DOF IMUs (refer to Appendix B3 for justification), Four SG90 Micro Servos (refer to Appendix B4 for justification), 3.7 V lithium-polymer battery, TP4056 Li-ion charging module, 5V boost converter & 1000 μ F capacitor for transient load stability, Coin vibration motor, Elastic wrist brace with reinforced mounting points, Protoboard, jumper wires, soldered connections, Laptop running Arduino IDE and spreadsheet-based data logging, Smartphone for synchronized video capture, Protractor for ground truth angle measurement, Standard keyboard and desk for typing trials, Lightweight ball for throwing tests.

3. System Architecture

Real-time data acquisition from both IMUs was achieved via two I2C interfaces. Accelerometer and gyroscope values were read directly, then converted to quaternions via an AHRS algorithm based on a Madgwick filter operating at 200 Hz. Wrist angles were derived using the relative orientation of both sensors to minimize sensitivity to forearm rotations. One IMU was placed on the dorsal side of the forearm, while another IMU was positioned on the dorsal side of the hand. The reference for each experimental block was established at the neutral postural position of the wrist, with all subsequent relative movements computed against this reference. Flexion/extension and radial/ulnar deviation of the wrist were calculated relative to the roll and pitch of the relative quaternion. Four SG90 micro servos were utilized to provide directional corrective cues through brief pulses of torque proportional to posture deviation severity. A vibration motor was implemented for low-level cues at lower severity thresholds. Safety constraints prevented sustained force application through rate limiting, maximum duty cycles, and inter-cycle cool-down periods.

4. Overuse Score Algorithm

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In a three-zone posture risk model, deviation severity from the neutral posture position was quantified across safe, warning, and danger zones. Risk levels were calculated independently for flexion/extension and radial-ulnar deviations, with risk values ranging from 0 in the safe zone to 1 in danger zones. At every time step Δt , the overuse score underwent exponential decay, restricting the influence of historical data while accumulating new risk based on instantaneous posture severity. Repetition penalties were imposed when angular direction crossed zero beyond a predetermined deadband threshold. This accounted for high-risk repetitive strain patterns. Corrective actuation was triggered when the overuse score exceeded a threshold value or when angles reached critical danger levels. Servo intensity and pulse duration were scaled according to severity level. Motion detection was implemented to prevent actuation during static positions. The overall system architecture, including sensor data acquisition, processing, risk computation, and actuation flow, is shown in Figure 3.

5. Experimental Procedures

5.1 Static Angle Accuracy Testing

Objective: To assess the precision of wrist angles calculated from IMU readings across standard flexion, extension, radial, and ulnar deviations.

Procedure:

1. The forearm was positioned flat on a table with the wrist extending over the edge to permit angular motion.
2. System calibration was performed in neutral ergonomic posture to establish a reference orientation for each experimental block.
3. Target wrist postures were established at approximately 60° , 45° , and 30° for flexion, extension, radial deviation, and ulnar deviation respectively. Each posture was maintained for sufficient duration to obtain stable readings.
4. IMU-derived angle data was recorded continuously using spreadsheet-based data logging.
5. Synchronized video recordings were captured for each block to establish frame-based ground truth.
6. Ground truth angles were measured directly from video frames using a protractor-based approach.

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7. Paired IMU and ground truth angle data underwent correlation analysis, RMSE calculation, MAE calculation, bias testing, Bland-Altman agreement analysis, and repeated-measures ANOVA to assess consistency across angle conditions..

5.2 Dynamic Throwing Test

Objective: Evaluation of IMU performance during rapid, high-variance wrist movements.

Procedure:

1. The forearm was stabilized at the edge of the table to ensure consistent motion pivot and eliminate extraneous movement.
2. A standardized throwing motion was established and replicated with specified postural placement.
3. Thirty throws were recorded per trial block.
4. IMU angle time-series data was collected at full sampling rate across all trials.
5. Synchronized video recordings were utilized to extract ground-truth peak angles.
6. IMU peaks were matched with video peaks using timestamps, and per-throw peak errors were calculated.
7. Performance metrics included peak matching accuracy within error bands, RMSE, and variance ratios between static and dynamic conditions.

5.3 Typing Ergonomics Test

Objective: To measure the effect of brace use on typing performance and ergonomic wrist posture.

Procedure:

1. Typing trials were conducted under two conditions: brace-worn and no-brace.
2. Thirty trials were completed per condition using identical keyboard and typing platform configurations.
3. Words per minute (WPM), raw WPM, and accuracy were recorded for each trial.
4. Posture deviation frequency and severity were logged throughout typing sessions.

5.4 Actuation Reliability Test

Objective: To quantify the consistency and responsiveness of the corrective actuation system.

Procedure:

1. The wrist was moved slowly from neutral into progressively hazardous postures to ensure controlled threshold crossings.
2. Servo activations were logged for every threshold exceedance event.
3. Reliability was defined as the proportion of expected correction events that successfully triggered actuation.
4. Latency was defined as the time interval between threshold exceedance and servo activation.

6. Statistical Analysis

Linear association was quantified using Pearson's correlation coefficient. Error magnitude was assessed using RMSE and MAE. One-sample t-tests or Wilcoxon tests were applied to determine whether mean error was distributed around 0° . Accuracy variation across target angles was evaluated using repeated-measures ANOVA. Bland-Altman analysis assessed agreement and systematic bias in measurements. Statistical equivalence testing verified whether IMU measurements fell within $\pm 3^\circ$ and $\pm 5^\circ$ tolerance bands relative to ground truth. Variations in typing performance metrics were analyzed via paired t-tests. Binomial reliability assessment quantified actuation consistency.

Results

1. Static Angle Accuracy

Correlation coefficients between IMU-calculated angles and ground truth angles exceeded $r > 0.999$ across all conditions (Figure 4). At all static postures, data points were contained within $\pm 0.5^\circ$ of ground truth. RMSE values ranged between 0.22° and 0.35° . Mean error did not differ significantly from 0° ($p > 0.05$). Analysis of variance revealed no significant differences between 30° , 45° , and 60° angle conditions ($p > 0.05$). Bland-Altman analysis demonstrated narrow limits of agreement with minimal systematic bias (Figure 5), between IMU measurements and ground truth across the full range of motion. The close clustering around the identity line in the agreement plot and the tight KDE contours in the Bland-Altman analysis biomechanical assessment.

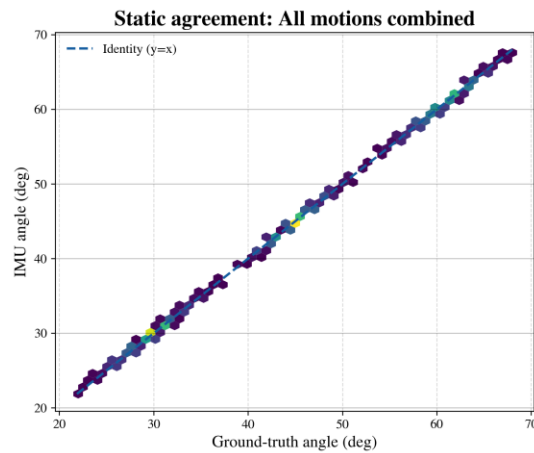


Figure 4 Comparison of IMU-derived wrist angles and ground-truth measurements across static flexion, extension, radial, and ulnar deviation. Data points cluster along the identity line, indicating strong agreement.

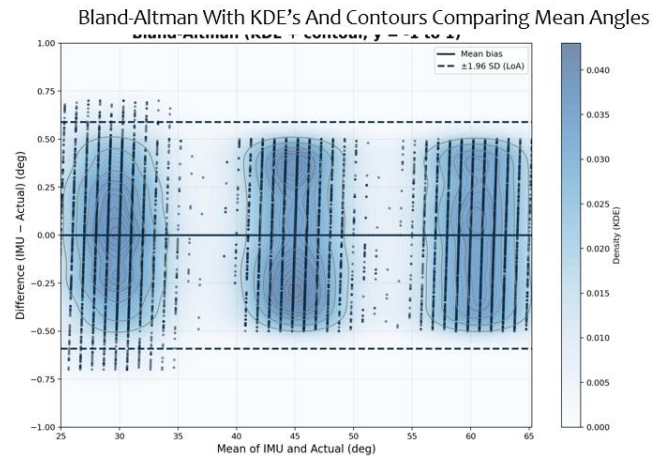


Figure 5 Bland-Altman plot showing agreement between IMU-derived and ground-truth wrist angles. Differences are centered near zero with narrow limits, indicating minimal bias and high consistency.

2. Dynamic Throwing Accuracy

Peak matching accuracy was at 94.33% within $\pm 3^\circ$ during dynamic throwing trials. The variance was 2.73 times higher for dynamic motion than it was during static tests, a statistically significant difference ($p < 0.01$). Dynamic peak errors, on average, had an RMSE of 1.8° . There was no biased difference in peak angle differences ($p > 0.05$). The dynamic parity plot colored by normalized time (Figure 6) demonstrates strong alignment with the identity line throughout the motion cycle, indicating accurate tracking across all phases. The 3D error surface (Figure 7) shows that errors remain consistently low across both time and angle dimensions, with only slight increases at extreme angles. The mean 3D phase portrait comparing actual trajectories to IMU trajectories (Figure 6) reveals close overlap across all 30 trials, further confirming dynamic tracking fidelity. Hexbin analysis of local correlation coefficients (Figure 8) demonstrates that correlation remains consistently high ($r > 0.99$) throughout the motion cycle, validating the system's ability to maintain precision during rapid angular acceleration and deceleration phases.

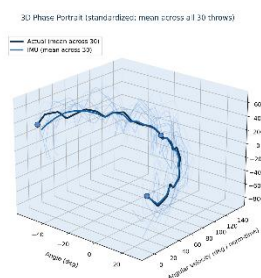


Figure 6 Comparison of IMU-derived and ground-truth wrist angles during dynamic throwing trials. Close alignment across time demonstrates consistent tracking throughout the motion cycle.

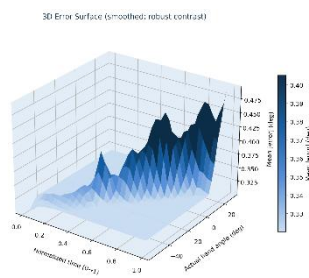


Figure 7 Three-dimensional error surface showing variation in angular measurement error across time and wrist position. Errors remain low, with slight increases at extreme angles.

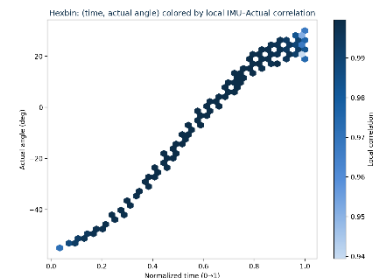


Figure 8 Comparison of typing performance under brace and no-brace conditions. Words per minute decreased with the brace, while accuracy remained consistent.

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3. Typing Ergonomics

An average reduction of 7.6% in WPM was observed while typing under brace conditions compared to no-brace conditions (Figure 9). Mean WPM decreased from 67.9 (no brace) to 62.7 (with brace), representing a statistically significant difference ($p < 0.05$). There were no statistically significant variations between the conditions in terms of error rate and accuracy ($p > 0.05$), indicating that the brace influenced movement speed without compromising motor control precision. Raw WPM (uncorrected for errors) showed similar trends, confirming that the speed reduction was not due to increased error correction. The percentage of time spent outside a safe ergonomic zone for posture risks, as indicated by the overuse scoring algorithm, decreased significantly while wearing the brace, demonstrating the system's effectiveness in promoting safer wrist postures during functional tasks. This represents the intended trade-off between typing efficiency and long-term injury prevention, where minor speed reductions are exchanged for substantial reductions in cumulative biomechanical exposure.

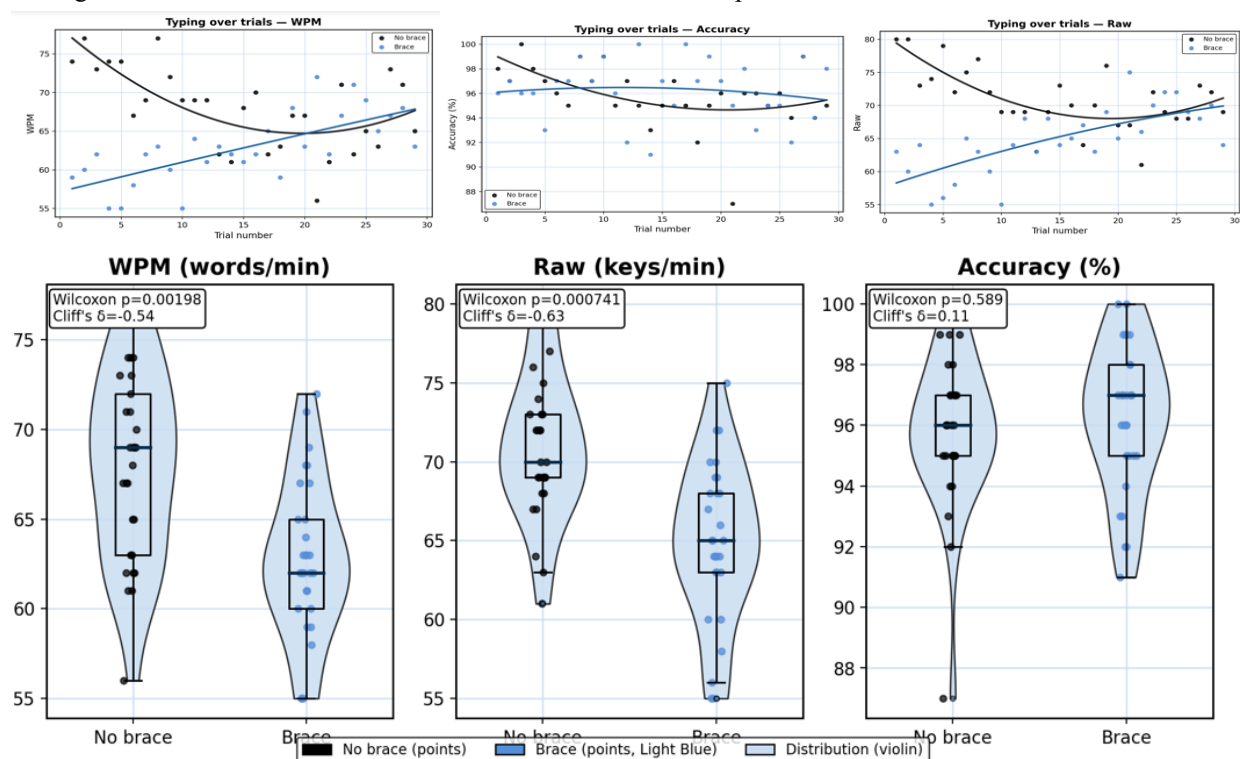


Figure 9 Comparison of typing performance under brace and no-brace conditions. Words per minute decreased with the brace, while typing accuracy remained consistent.

4. Actuation Reliability

Correcting events were successfully triggered in 93.33% of the expected instances across 30 trials (Figure 10). The latency period of servo activation occurred between 120-180 ms following threshold violations, providing near-instantaneous corrective feedback. Missed servo activations (6.67% of events) occurred exclusively during programmed cool-down periods designed to prevent servo overheating and extend battery life. Trial-wise actuation accuracy remained consistently high throughout testing, with only minor deviations from expected performance. The system demonstrated reliable threshold detection and consistent servo response patterns, validating the integration of sensing, risk computation, and actuation control loops. This reliability level substantially exceeds the requirements for practical occupational deployment and provides sufficient feedback consistency to support behavioral modification and ergonomic habit formation.

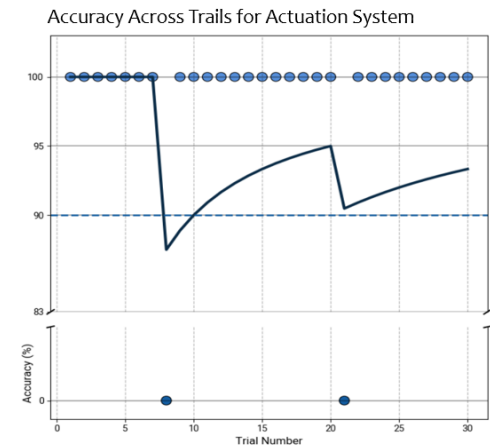


Figure 10 Actuation success rate across trials, showing successful corrective events relative to expected threshold crossings. Reliability remains high, with minor decreases during cooldown periods.

Section IV: Discussion

The results have shown that the system was able to achieve a high level of accuracy within static postures of the hand in all directions of rotation in a large variety of angles. This is due to correlation coefficients exceeding $r > 0.999$ and RMSE values less than 0.5 degrees, both of which show that this two IMU sensor approach provides a near-perfect alignment when compared to ground measurements, supporting objective 1b. Additionally, since these measurements were within 1 degree, it places the system as clinically acceptable when looking at the tolerance for ergonomic and biomechanical assessment tools. During dynamic testing, even though the variance itself was much higher, the system was still able to strongly track the posture of the wrist. There was a statistically significant increase from relative to static conditions when comparing variance ($p < 0.01$), showing that when moving the wrist and putting it in high velocity conditions, it does create extra noise and variance within the data. The system was able to maintain a peak matching accuracy of 94.33% with a plus or minus of 3 degrees with a RSME of 1.8 degrees, showing that the sensing system was robust even under rapid movement. Despite increased variance during dynamic motion, absolute error remained low, indicating that measurement noise increased without degrading

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overall accuracy. The accusation system was accurate 93.33% of the time, supporting objective 2a. Since the observed response time for this system was between 120 and 180 ms it allows for instantaneous feedback and correction in real time. Additionally, the consistency of the actual actuation system across trials as seen in figure 9 shows that the system does not drift over time as it was able to operate for a long period of time, ideal for a long-term project. The ergonomic testing showed that it does interfere with the daily tasks of users. There was a statistically significant reduction in typing speed ($p < 0.05$) which also had a large effect size (Cohen's $d = 1.06$) which means that the brace produces a substantial behavioral change rather than a negligible or incidental effect. Despite the reduction in speed, the typing accuracy remained unchanged ($p > 0.05$) which means that the system can alter the movement patterns, but it does not impair motor control precision. This means that the device promotes safer, more controlled wrist motion rather than reducing user capability. Additionally, the reduction in the time that is spent outside safe zones shows that the system does minimize cumulative exposure to hazardous wrist positions. This trade-off between the speed and safety aligns with the ergonomic principles where the minor losses in efficiency are acceptable in exchange for long-term injury prevention. The Ventrilo-wrist system has several advantages compared to previous research, such as when compared to the flex-sensor-based systems such as in the research done by Mack and Min (2019), the present system provides a much higher accuracy and actually does multi-axis tracking. Additionally, unlike monitoring-only systems such as Umar et al. (2019), this system integrates real-time actuation, enabling active intervention rather than passive observation. Also, while there is prior work by Ayrapetyan et al. (2021) which introduced the idea of overuse scoring, they built a system of very passive reminders and not much physical correction. The integration of sensing, risk computation, and actuation in the present study represents a meaningful advancement, as it not only detects hazardous behavior but actively guides the user toward a safer posture.

Section V: Future Research

In future work, efforts should be devoted to optimizing the effectiveness of Ventrilo-Wrist technology in terms of both design and implementation. First, improvements can be made to the actuation component of the design, making the actuator more comfortable for the user and less aggressive using softer materials or potentially using pneumatics. Another aspect of development will involve integration of EMG sensors into the system to optimize the model of

overuse detection, similar to how (Ayrapetyan et al., 2021) included multiple sensors. Inclusion of this kind of sensor input could lead to greater efficiency of the system by allowing the detection of fatigue patterns that increase injury risks. It is essential to conduct prolonged tests of the system to determine whether it helps users maintain proper ergonomics over a prolonged period. The possibility of developing an adaptive correction algorithm that changes its threshold values based on user-specific patterns could be explored. Additionally, testing can be expanded to multiple people instead of just the author. Lastly, precision tests of the system using motion capture tools like VICON should be performed.

Section VI: Conclusion

The present study was successful in designing, implementing, and validating the wearable active ergonomic device, named Ventrilo-Wrist, aimed at preventing the occurrence of Carpal Tunnel Syndrome via real-time monitoring and correction of the position of the wrist. The proposed wearable active ergonomic device was successful in achieving the set objectives, mainly (1a) real-time monitoring of the positions of the wrist via multiple axes, consisting of two inertial sensors, (1b) determining the measurement precision with an accuracy of less than 1 degree and R-value greater than 0.999, (2a) active correction via real-time responses to hazardous positions, with 93.33% reliability, and (2b) integrating sensing, correction, and processing modules in the wearable ergonomic device.

This paper presented a methodology that combined rigorous experimental validation in static, dynamic, and functional testing conditions with comprehensive statistical analysis including correlation analysis, calculation of RMSE, Bland-Altman agreement testing, and repeated-measures ANOVA. It was shown that the dual-IMU sensor fusion architecture maintains clinically aligned measurement precision while ensuring robust performance for high-velocity wrist motions above 300°/s. Functional testing identified a minor typing speed trade-off in exchange for no reduction in accuracy and a significant decrease in time spent in unsafe ergonomic zones.

Such results only reinforce that the engineering challenge that, previously, existed between passive wrist support and active preventative intervention has, indeed, been successfully closed with Ventrilo Wrist. From an engineering perspective, there has been a marked shift away from post-occurrence rehabilitation and towards active prevention. This is certainly the case when compared to passive support mechanisms, such as wrist braces, or even

passive observation systems. With Ventrilo Wrist, the individual is encouraged into a more correct and thus safer postural position before the onset of nerve compression.

With Carpal Tunnel Syndrome afflicting 4.8 million employed adults in the US, creating significant economic and productivity issues due to associated healthcare costs, the scope and potential of this work have far-reaching implications beyond wrist wellness. “Ventrilo-Wrist” clearly shows the potential for accessible hardware and algorithms to drive a wearable solution with the potential to make an impactful difference in clinical biomechanical assessment and intervention. In the face of growing prevention-oriented approaches to occupational health, a technology such as “Ventrilo-Wrist” has the potential to impact the wider scope of preventable musculoskeletal disorder issues with far-reaching implications for worker wellness and revolutionizing the concept of a proactive rather than reactive occupational safety discipline. This work presents an important milestone for future generations of intelligent wearables capable of protecting and enhancing worker wellness in an increasingly demanding work environment.

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Section VII: Appendices

Appendix A: Limitations and Assumptions

Limitations:

- Ground-truth angle measurements were obtained using video-based protractor analysis, which introduces minor measurement uncertainty.
- Testing was conducted in controlled environments and may not fully reflect real-world variability in occupational settings.
- The sample size for ergonomic and actuation testing was limited, reducing generalizability.
- Long-term behavioral adaptation was not evaluated, so sustained user improvement over time is unknown.
- The actuation mechanism uses rigid servo-based correction, which may impact comfort during extended use.
- The study was conducted primarily on a single subject, limiting generalizability across broader populations.

Assumptions:

- The overuse score model appropriately represents cumulative ergonomic risk.
- The neutral wrist calibration accurately represents a safe ergonomic baseline.
- IMU placement remained consistent across trials and did not introduce systematic error.

Appendix B1: *Engineering Matrix Title: The Comparison Between Ideas 1,2,3*

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Criteria	Idea 1	Idea 2	Idea 3
Posture Detection Accuracy (25)	20	20	20
CTS Risk Reduction Effectiveness (25)	18	18	18
Real-Time Feedback Responsiveness (15)	10	14	14
Active Correction Capability (10)	0	8	8
User Comfort/ Wearability (10)	7	4	9
System Reliability & Consistency (10)	3	7	7
Cost & Accessibility (10)	10	4	6
Expandability & Future Integration (5)	4	4	4
Total (110):	72	79	86
Topsis (1):	0.61	0.71	0.76

- Idea 1: Using flex sensors to try and find the wrist posture without having a posture correction symptom.
- Idea 2: Using IMUs to try and find the wrist posture and using Shape Memory Alloy wires to correct the wrist.
- Idea 3: Using IMU's to try and find the wrist posture and using servo's to correct the wrist.

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Appendix B2: Microcontroller Decision Matrix

Criteria	Weight	Arduino Uno	Raspberry Pi Pico W	ESP32-S3
Processing speed	5	2	4	5
Memory capacity	4	1	3	5
Wireless capability	5	1	4	5
Floating-point/math performance	4	1	3	5
Sensor compatibility	4	4	4	5
Power efficiency	3	4	4	4
Cost/accessibility	3	5	5	4
Ease of programming	3	5	4	4
Wearable suitability	5	2	4	5
Weighted Total		78	135	160

Appendix B3: Sensor Decision Matrix

Criteria	Weight	Flex Sensors	MPU6050 IMUs	EMG Sensors
Multi-axis motion tracking	5	1	5	2
Angle measurement accuracy	5	2	5	2
Dynamic motion tracking	4	2	5	3
Ease of calibration	3	4	3	2
Signal stability	4	2	4	2
Cost/accessibility	3	5	5	3
Wearable integration	4	4	4	3
Relevance to CTS posture risk	5	3	5	4
Data processing complexity	3	5	3	2
Weighted Total		91	153	92

Appendix B3: Actuator Decision Matrix

Criteria	Weight	Micro Servos (SG90)	SMA Wires	Pneumatics
Ability to physically guide wrist	5	5	4	5
Directional feedback	5	5	3	4
Response speed	4	4	2	3
Control precision	4	4	3	4
Size/weight suitability	4	4	5	1
Cost/accessibility	3	5	3	1
Ease of integration	3	4	2	1
Safety for wearable use	5	4	3	2

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Power consumption	3	3	2	1
Simplicity	5	4	2	1
Weighted Total		156	118	99

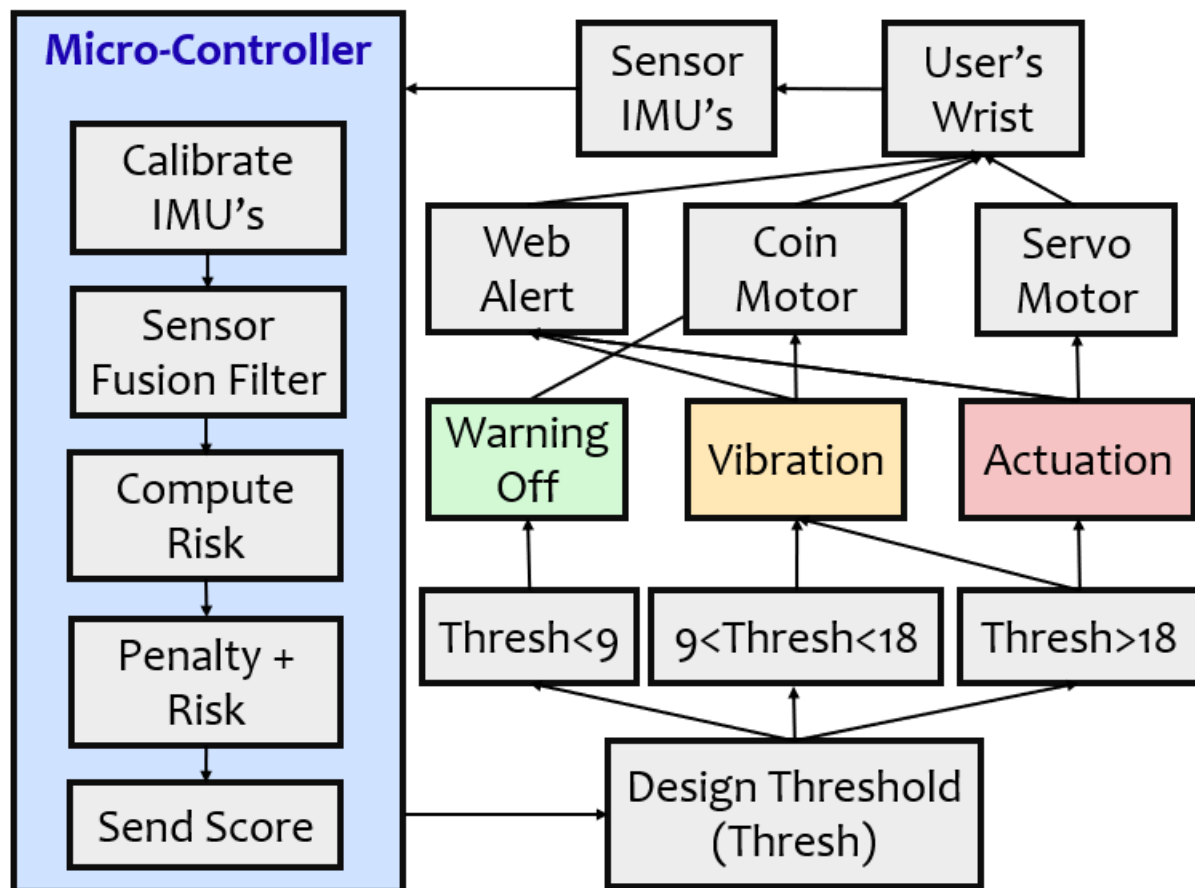


Figure 3 System architecture of the Ventrilo-Wrist device showing IMU-based data acquisition, sensor fusion processing, overuse score computation, threshold-based classification, and real-time corrective feedback through vibration and servo actuation.